Population Proxy Mapping Using Night-Time Lights

A Complete Research Diary

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Dataset: VIIRS VNL V2.1 Annual 2019 | Study areas: India, Nigeria

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1. What This Project Is About

The core idea is simple: at night, places where people live glow. Satellites can see this glow. So in theory, the brightness of night-time lights (NTL) over a region should tell you something about how many people live there. This project tests exactly that — can we use VIIRS satellite radiance data to predict population at the district level?

The dataset used is the VIIRS VNL V2.1 Annual Composite for 2019, downloaded from the Earth Observation Group (EOG) at Colorado School of Mines. Specifically the average_masked variant, which has background (non-lit areas), fires, and outliers already removed. This is the standard dataset used in published NTL-population literature.

The ground truth population data comes from WorldPop 2019 — a gridded population dataset at 1km resolution, built from census data and demographic modelling.

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2. The Dataset

2.1 VIIRS VNL V2.1

Full name: VNL_v21_npp_2019_global_vcmslcfg_c202205302300.average_masked.dat.tif

Downloaded from: https://eogdata.mines.edu/nighttime_light/annual/v22/2019/

Property	Value
Sensor	VIIRS DNB onboard Suomi-NPP
Year	2019
Version	V2.1 (latest at time of download)
Resolution	15 arc-second (~500m at equator)

Property	Value
Unit	nW/cm²/sr (radiance)
CRS	EPSG:4326
Coverage	Global (180W to 180E, 75N to 65S)
File size	~295 MB (average_masked variant)
Key property	Background zeroed out, fires removed, outliers removed

2.2 Why average_masked and not the others

The EOG directory contains several variants. Here is what each one is and why we did not use them:

- average — raw average radiance, includes background noise. Needs manual cleaning.
- average_masked — same as average but with background zeroed and outliers removed. This is what we used.
- median / median_masked — more robust to outliers but non-standard for population work.
- max, min — not useful for population estimation.
- cf_cvg — cloud-free coverage count, a quality layer. Not radiance data.

2.3 WorldPop 2019

Source: https://data.worldpop.org — UN-adjusted, 1km resolution gridded population.

Downloaded separately for India (ind_ppp_2019_1km_Aggregated.tif) and Nigeria (nga_ppp_2019_1km_Aggregated.tif). These give pixel-level population counts that we aggregate to district level using zonal statistics.

2.4 Admin boundaries

GADM v4.1 Level 2 shapefiles from https://gadm.org — district level for both countries. India has 676 districts, Nigeria has 775 LGAs.

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3. The Pipeline (What the Code Does)

Everything runs in Google Colab on CPU. The full pipeline in order:

Step 1 — Mount Drive and load the TIF

The global TIF was uploaded to Google Drive and mounted in Colab. Rasterio reads it directly without loading the whole file into RAM.

Step 2 — Download boundaries and WorldPop

GADM shapefiles downloaded via urllib. WorldPop TIF downloaded via requests with streaming to avoid memory issues.

Step 3 — Zonal statistics

rasterstats.zonal_stats() computes NTL sum, mean, and max per district polygon by overlaying the shapefile on the raster. Same process repeated for WorldPop to get population per district. This produces one row per district with NTL features and population ground truth.

Step 4 — Feature engineering

Both NTL and population follow power-law distributions — a few bright/dense districts and many dim/sparse ones. Log-transform (log1p) linearises this relationship and is standard in the literature.

Features created:

- log_ntl_sum — log of total radiance summed over all pixels in district
- log_ntl_mean — log of mean radiance per pixel
- log_ntl_max — log of brightest single pixel
- log_area — log of district area in km^2 (added as control variable)
- lit_fraction — mean/std ratio of radiance, proxy for how uniformly distributed light is within the district

Step 5 — Urban/rural classification

KMeans clustering (k=2) on log_ntl_sum and log_ntl_mean assigns each district to urban or rural. The cluster with higher mean NTL is labelled urban=1. This is more principled than an arbitrary quantile threshold.

India result: ~170 urban, ~500 rural districts.

Step 6 — Models

Two models tested: Linear Regression and Random Forest. Both fitted on log-transformed features predicting log_pop. Evaluation via 5-fold cross-validation with R^2 as the metric.

Step 7 — Stratified model (main contribution)

Instead of one global model, fit separate Linear Regression models for urban and rural districts. Prediction for each test district uses whichever sub-model matches its urban/rural label. Improvement tested for statistical significance using paired t-test across fold scores.

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4. India Results

4.1 Core model results

Model	CV R^2	Std	Delta	p-value
Baseline NTL (3 features)	0.8289	0.0270	—	—
+ Area control	0.8279	0.0234	-0.0001	n/a
+ Urban/rural stratification	0.8582	0.0195	+0.0293	0.0267 ✓
+ NDVI (stratified)	0.8579	0.0210	-0.0003	n/s

4.2 What each result means

Baseline $R^2 = 0.83$

NTL alone explains 83% of variance in district-level population across India. This is competitive with published results. ntl_sum is the strongest single predictor (r=0.822 with population).

Area control: no effect

A key concern in NTL-population models is that ntl_sum might just be a proxy for district size (bigger districts = more total light = more people). We explicitly tested this:

- Correlation of area vs ntl_sum: r = 0.119 (near zero)
- Correlation of area vs population: r = 0.065 (near zero)
- Adding log_area as a feature: $\Delta R^2 = -0.0001$

Conclusion: district size is NOT a confound. ntl_sum genuinely measures light intensity, not just area. This is worth stating explicitly in the paper because most published work does not run this check.

Stratification: $+0.03\ R^2$, $p = 0.027$

The urban/rural split is the main contribution. Fitting separate models for urban and rural districts improves CV R^2 from 0.8289 to 0.8582, a gain of +0.0293. The paired t-test across 5 folds gives p=0.0267, confirming this is statistically significant and not a random fluctuation.

Why does stratification help? Urban districts have a different NTL-population relationship than rural ones. In dense urban cores, NTL saturates (sensor hits its upper limit) and underestimates true population. In rural areas, NTL is sparse but population can still be high (agricultural districts). A single global model cannot capture both regimes simultaneously.

NDVI: failed, and why that is interesting

We downloaded MODIS NDVI data and added mean district NDVI as a covariate, hypothesising that vegetation density could help correct for rural areas where NTL underestimates population. The result was negative: $\Delta R^2 = -0.0003$, no improvement.

Looking at the residual map explains why. The most under-illuminated districts (highest positive residuals = model underestimates population) are:

- Malappuram (residual: +1.60, NDVI: 0.74, pop: 4.3M)
- Kannur (residual: +1.46, NDVI: 0.66, pop: 2.5M)
- Kozhikode (residual: +1.41, NDVI: 0.71, pop: 3.2M)
- Wayanad (residual: +1.21, NDVI: 0.77, pop: 0.8M)

All of these are Kerala districts. Kerala has dense forest cover AND high population density, but a dispersed rural settlement pattern with low commercial lighting. High NDVI here co-occurs with high population — so adding NDVI as a feature actively confounds the model rather than helping it.

This is a geographically meaningful negative result. NDVI correction for NTL-population models is not universally applicable and fails in tropical regions with dispersed settlements.

4.3 Most over-illuminated districts

The most over-illuminated districts (model overestimates population) include:

- Kinnaur (residual: -3.00) — high-altitude Himachal Pradesh, border area, possible military infrastructure lighting
- Leh/Ladakh (residual: -1.14) — sparse population, high NTL possibly from military/strategic installations
- Gandhinagar (residual: -1.04) — state capital, high commercial/government lighting relative to residential population
- Morbi (residual: -1.00) — industrial city (ceramics/tiles industry), high industrial NTL

This pattern is interpretable: the model fails where NTL source is non-residential. Industrial zones, border infrastructure, and government campuses produce light that does not correspond to population.

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5. Nigeria Results

5.1 Results

Model	CV R^2	Std	Delta	p-value
Baseline NTL (3 features)	0.1676	0.0582	—	—
+ Urban/rural stratification	0.2511	0.0693	+0.0835	0.0039 ✓

5.2 Why Nigeria R^2 = 0.17 is correct, not a bug

The low baseline R^2 in Nigeria is real and expected. Three structural factors destroy the NTL-population correlation there:

- Gas flaring in the Niger Delta: oil extraction produces massive radiance in the delta region (Rivers, Bayelsa, Delta states) with essentially zero residential population. These pixels are not removed by the average_masked filter because they are persistent, not ephemeral.
- Sensor saturation in Lagos: Lagos LGA has ~15 million people in a tiny area. The VIIRS sensor saturates — it maxes out — meaning the recorded radiance is lower than the true radiance. NTL therefore systematically underestimates the most densely populated area in West Africa.
- Low rural electrification in the north: states like Sokoto, Zamfara, Yobe, Borno have large populations with very limited electricity access. NTL is near zero but population is real and substantial.

These three factors pull the NTL-population correlation in different directions across the country and collectively explain why a single regression achieves only R^2 = 0.17.

5.3 Why stratification still helps (p = 0.004)

Despite the low absolute R^2, stratification still improves performance significantly: +0.08 R^2, p=0.004. This is actually a stronger p-value than India (0.004 vs 0.027), which means the urban/rural distinction is even more important in Nigeria than India.

The reason is that urban Nigerian districts (Lagos, Kano, Ibadan, Abuja) have a fundamentally different NTL-population regime than rural ones. Separating them allows each sub-model to find a better fit within its own regime, even if neither regime fits well in absolute terms.

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6. Cross-Country Comparison

Country	n (districts)	Baseline R^2	Stratified R^2	Delta	p-value	Key context
India	670	0.8289	0.8582	+0.0293	0.0267 ✓	High electrification, diverse geography
Nigeria	695	0.1676	0.2511	+0.0835	0.0039 ✓	Gas flares, saturation, low rural electrification

The cross-country comparison is the most important result in the paper. Two findings:

- Stratification consistently improves NTL-population models regardless of country or development context ($p<0.05$ in both cases).
- Absolute model accuracy depends heavily on structural factors like electrification rate and NTL contamination sources, varying from $R^2=0.83$ in India to $R^2=0.17$ in Nigeria.

This means the paper has two distinct contributions: a methodological one (stratification helps universally) and an empirical one (NTL reliability is context-dependent in predictable ways).

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7. What Failed and Why

IIIT Dharwad campus prediction

Early in the project, GEE was used to extract NTL values for a 280m buffer around the IIIT Dharwad campus coordinates (75.024751, 15.392891). The extracted values were:

- avg_rad_sum: 3.54 | avg_rad_mean: 3.46 | avg_rad_max: 4.85

These were fed into the trained India model and produced a population prediction. The result appeared reasonable but was meaningless. The model was trained on district-level data (units of hundreds of km² and millions of people). Feeding it a 300m buffer with 1-2 pixels of NTL data is a scale mismatch — the model has no valid basis for extrapolation to that resolution. Any result is coincidental.

Bangladesh attempt

Bangladesh was tried as a second validation country. It failed for a simple reason: Bangladesh has only 64 districts at GADM Level 2. With 5-fold CV, each fold has ~12 districts, then split urban/rural that is ~6 per group — far too few to train a stable model. Results showed CV std = 0.25 (massive variance) and p = 0.29. This is a sample size problem, not a real negative result.

NDVI as covariate

Already covered in Section 4.2. Short version: NDVI adds no value and slightly hurts the model in India because of the Kerala confound. Not tested in Nigeria.

Area normalisation

Dividing ntl_sum by district area (ntl_density) caused R^2 to collapse from 0.83 to 0.47. Initial concern was that ntl_sum might be a size proxy. Correlation analysis showed this was not the case (r=0.119 for area vs ntl_sum). The collapse happened because area normalisation destroys the signal — districts in India are drawn roughly by population, not by area, so raw ntl_sum is the correct feature. Lesson: check correlations before normalising.

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8. The Paper: How to Frame It

8.1 Suggested title

Urban-Rural Stratification Consistently Improves VIIRS-Based Population Proxy Mapping: Evidence from India and Nigeria

8.2 Abstract numbers to use

- India baseline $R^2 = 0.83$, stratified $R^2 = 0.86$, $\Delta = +0.03$, $p = 0.027$
- Nigeria baseline $R^2 = 0.17$, stratified $R^2 = 0.25$, $\Delta = +0.08$, $p = 0.004$
- Size confound explicitly tested and ruled out ($r = 0.119$)
- NDVI tested and rejected with geographic explanation
- 670 Indian districts + 695 Nigerian LGAs, both 5-fold CV

8.3 Paper structure (4 pages, IGARSS format)

- Section 1 — Introduction: NTL as population proxy, gap in literature (single-regime models, no confound testing, no cross-country validation)
- Section 2 — Data and Methods: VIIRS VNL V2.1, WorldPop 2019, GADM Level 2, zonal statistics pipeline, KMeans stratification, 5-fold CV
- Section 3 — Results: Table with all model variants for both countries, residual map for India, choropleth maps
- Section 4 — Discussion: Why stratification works, why Nigeria fails at baseline, Kerala dispersed settlement explanation for NDVI failure, size confound analysis
- Section 5 — Conclusion: Stratification is universally applicable, NTL reliability is context-dependent

8.4 Target venue

IGARSS 2026 (IEEE International Geoscience and Remote Sensing Symposium). 4-page limit. This is the premier conference in remote sensing. Submission deadline typically January-February. Review is fast (~6 weeks). Acceptance rate ~50% for well-executed methodology papers.

Backup: ISPRS Congress 2026, or Remote Sensing Letters (journal, 4-page short letter format).

8.5 Key citations you will need

- Elvidge et al. 2017 — VIIRS night-time lights (Int. J. Remote Sensing) — cite for dataset
- Elvidge et al. 2021 — Annual time series VNL V2 (Remote Sensing 13(5):922) — cite for V2 methodology
- Lo 2001 or Sutton et al. 2001 — early NTL-population work — cite as baseline comparison
- Worldpop paper (Tatem et al.) — cite for ground truth data
- Any paper using stratified or disaggregated NTL models — cite as related work gap you are filling

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9. Files and Code Location

File	Location
NTL raster	Google Drive: NTL_project/VNL_v21_npp_2019_global_vcmslcfg_c202205302300.average_masked.dat.tif
India population	Google Drive: NTL_project/ind_pop_2019.tif
Nigeria population	/content/nga_pop_2019.tif (re-download needed)
India boundaries	Downloaded fresh: GADM v4.1 IND Level 2
Nigeria boundaries	Downloaded fresh: GADM v4.1 NGA Level 2
Results CSV	Google Drive: NTL_project/results_final.csv
India choropleth	Google Drive: NTL_project/choropleth_india.png
Nigeria choropleth	Google Drive: NTL_project/choropleth_nigeria.png
Residual comparison map	/content/residual_comparison.png
Full notebook	Google Colab — re-run from top, all cells in order

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10. Honest Limitations for the Paper

Write these in the discussion section. Do not hide them — reviewers will find them anyway and it is better to preempt.

- Single year (2019): temporal generalisability is not tested. Results may differ for other years if NTL patterns shift.
- Only two countries: India and Nigeria are not a representative global sample. Both are large, populous countries. Results may not hold for small island states, high-latitude countries, or conflict zones.
- LR beats RF: Random Forest did not outperform Linear Regression. This is because with only 3-5 features, RF overfits folds. With more features (e.g. temporal stack, land cover, road density) RF would likely win.
- Lit fraction proxy is approximate: the mean/std ratio used as lit_fraction is not a true lit pixel count. A proper implementation would threshold pixels above a radiance cutoff, but this was too slow on the global raster without tiling.
- Gas flare removal incomplete: the average_masked filter removes ephemeral fires but not persistent gas flares. This is a known limitation of VNL V2.1 and is documented in Elvidge et al. 2021.
- Urban/rural binary is a simplification: KMeans with k=2 is principled but a spectrum of urbanisation exists. Peri-urban districts are forced into one category.

End of Research Diary

When you sit down to write the paper, start with Section 8. Everything else is your Methods and Results sections already written.